

RESULTS

Results

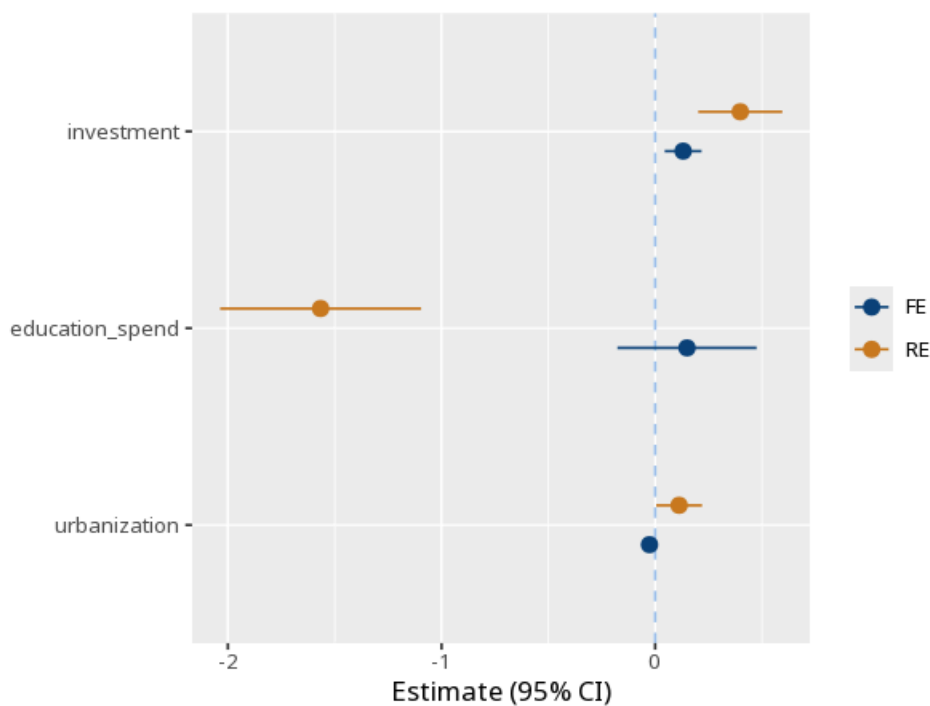
01 · HAUSMAN TEST

fixed vs. random effects?

Table 1. Hausman test

	Fixed effects	Random effects	Difference
investment	0.13 (0.04) [0.04, 0.22]	0.40 (0.10) [0.20, 0.59]	-0.27
education_spend	0.15 (0.16) [-0.18, 0.47]	-1.57 (0.24) [-2.04, -1.10]	1.72
urbanization	-0.03 (0.02) [-0.06, 0.01]	0.11 (0.05) [0.00, 0.22]	-0.14
Num.Obs.	100		
N entities	10		
R²	.44	.93	
Hausman $\chi^2(3) = 572.56, p < .001$			

Figure. FE vs. RE



Understanding the numbers

Fixed effects. The fixed-effects estimate for this predictor - consistent even if the random-effects assumption fails. Here, the fixed-effects estimate is 0.13.

Random effects. The random-effects estimate for the same predictor - more efficient, but only valid if entity effects are uncorrelated with the predictors. Here, the random-effects estimate is 0.40.

Difference. The difference between the fixed- and random-effects estimates - the quantity the Hausman test evaluates. Here, the difference is -0.27 .

N. The total number of entity-period observations underlying both models. Here, $N = 100$.

N entities. The number of distinct entities in the panel. Here, $N \text{ entities} = 10$.

R². Each model's own R², reported alongside the Hausman comparison. Here, R² is FE .44 / RE .93.

Hausman χ^2 . The Hausman test statistic - how far the fixed- and random-effects estimates diverge, in aggregate across predictors. Here, $\chi^2(3) = 572.56$, $p < .001$.

How to read this test

Tests whether the random-effects assumption holds. A p below alpha means the estimates differ systematically $\rightarrow$ use fixed effects; a non-significant p means random effects is acceptable (and more efficient). Method: R plm package (Croissant & Millo, 2008). The standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Your significance threshold (α) is 0.05.

APA template: A Hausman test comparing the fixed- and random-effects estimates gave $\chi^2(3)=572.56$, $p < .001$.

Statistical basis: Hausman specification test (Hausman, J. A. (1978). "Specification tests in econometrics." *Econometrica*, 46(6), 1251-1271.); Implementation (plm package) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." *Journal of Statistical Software*, 27(2), 1-43.). Full references in the exported CITATIONS.txt.

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